

● MEDIUM

● ADVANCED MATH

Nonlinear functions

10 questions · ~15 min

06

Q1

ADVANCED MATH

The function f is defined by $f(x) = 7x^3$. In the xy -plane, the graph of $y = g(x)$ is the result of shifting the graph of $y = f(x)$ down 2 units. Which equation defines function g ?

A. $g(x) = \frac{7}{2}x^3$

B. $g(x) = 7x^{\frac{3}{2}}$

C. $g(x) = 7x^3 + 2$

D. $g(x) = 7x^3 - 2$

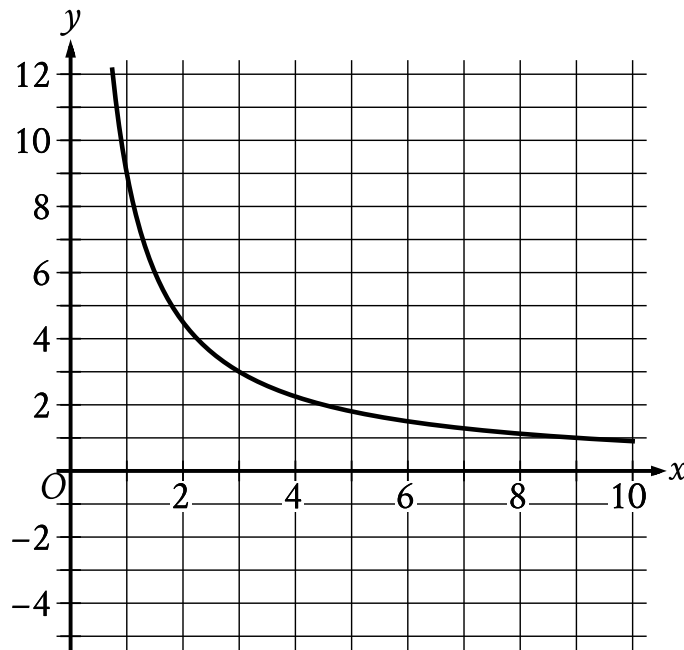
From 2005 through 2014, the number of music CDs sold in the United States declined each year by approximately 15% of the number sold the preceding year. In 2005, approximately 600 million CDs were sold in the United States. Of the following, which best models C , the number of millions of CDs sold in the United States, t years after 2005?

A. $C = 600(0.15)^t$

B. $C = 600(0.85)^t$

C. $C = 600(1.15)^t$

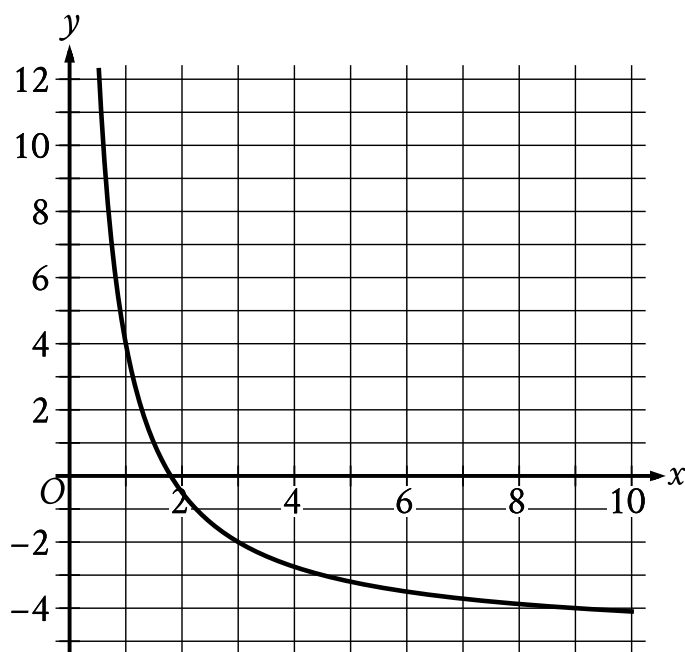
D. $C = 600(1.85)^t$



- Moving from left to right:
 - The curve is shown in quadrant 1.
 - The curve trends down sharply to x equals 3.
 - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
 - (2 comma nine halves)
 - (3 comma 3)
 - (6 comma three halves)

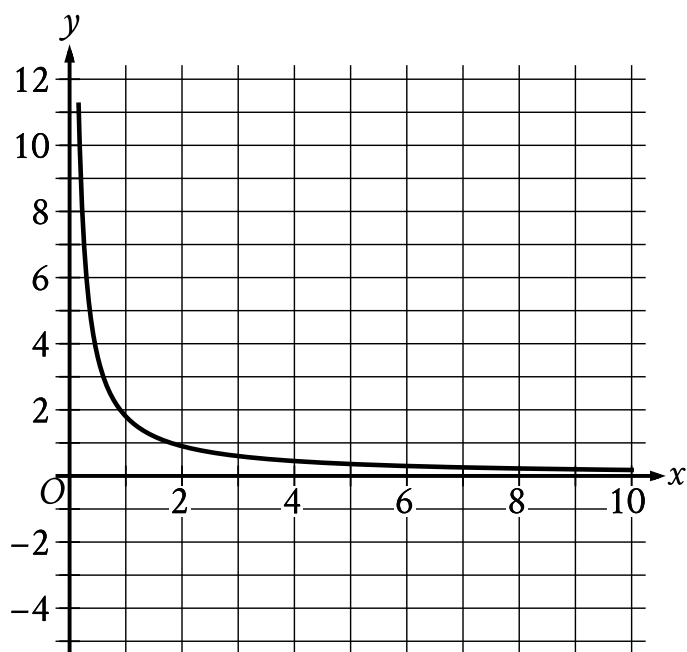
The graph of the rational function f is shown, where $y = f(x)$ and $x \geq 0$. Which of the following is the graph of $y = f(x) + 5$, where $x \geq 0$?

A.



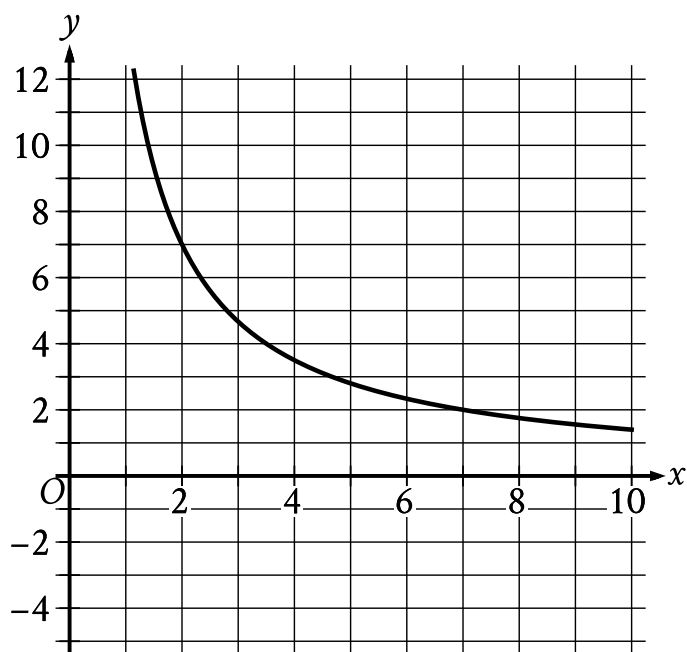
- Moving from left to right:
 - The curve passes from quadrant 1 to quadrant 4.
 - The curve trends down sharply to x equals 3.
 - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals negative 5.
- The curve passes through the following points:
 - (2 comma negative one half)
 - (3 comma negative 2)
 - (6 comma negative seven halves)

B.



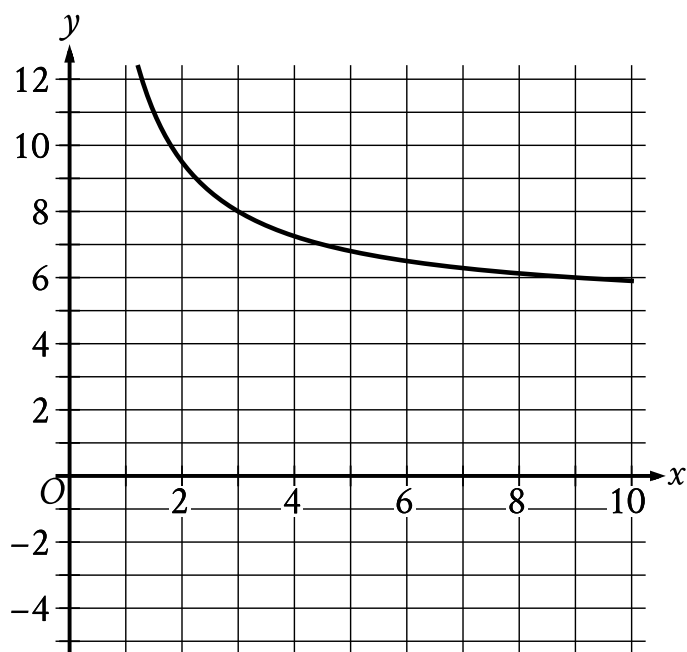
- Moving from left to right:
 - The curve is shown in quadrant 1.
 - The curve trends down sharply to x equals 1.
 - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
 - (2 comma nine tenths)
 - (3 comma three fifths)
 - (6 comma three tenths)

C.



- Moving from left to right:
 - The curve is shown in quadrant 1.
 - The curve trends down sharply to x equals 2.
 - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 0.
- The curve passes through the following points:
 - (2, 7)
 - (3, $\frac{14}{3}$)
 - (6, $\frac{7}{3}$)

D.



- Moving from left to right:
 - The curve is shown in quadrant 1.
 - The curve trends down sharply to x equals 3.
 - The curve then trends down gradually.
- As x increases, the curve approaches the line y equals 5.
- The curve passes through the following points:
 - $(2, \frac{19}{2})$
 - $(3, 8)$
 - $(6, \frac{13}{2})$

Q4

ADVANCED MATH

The function $f(t) = 40,000 \cdot 2^{\frac{t}{790}}$ gives the number of bacteria in a population t minutes after an initial observation. How much time, in minutes, does it take for the number of bacteria in the population to double?

- A. 2
- B. 790
- C. 1,580
- D. 40,000

Q5

ADVANCED MATH

x	24	30	32
$f(x)$	-8	-8	8

For the quadratic function f , the table shows three values of x and their corresponding values of $f(x)$. Which equation defines f ?

- A. $f(x) = (x - 24)(x - 30) + 4$
- B. $f(x) = (x - 24)(x - 30) - 8$
- C. $f(x) = (x - 8)(x - 32) + 32$
- D. $f(x) = (x - 8)(x - 32) - 32$

The equation $E(t) = 51.8^t$ gives the estimated number of employees at a restaurant, where t is the number of years since the restaurant opened. Which of the following is the best interpretation of the number 5 in this context?

- A. The estimated number of employees when the restaurant opened
- B. The increase in the estimated number of employees each year
- C. The number of years the restaurant has been open
- D. The percent increase in the estimated number of employees each year

Immanuel purchased a certain rare coin on January 1. The function $f(x) = 651.03^x$, where $0 \leq x \leq 10$, gives the predicted value, in dollars, of the rare coin x years after Immanuel purchased it. What is the best interpretation of the statement “ $f(8)$ is approximately equal to 82” in this context?

- A. When the rare coin's predicted value is approximately 82 dollars, it is 8% greater than the predicted value, in dollars, on January 1 of the previous year.
- B. When the rare coin's predicted value is approximately 82 dollars, it is 8 times the predicted value, in dollars, on January 1 of the previous year.
- C. From the day Immanuel purchased the rare coin to 8 years after Immanuel purchased the coin, its predicted value increased by a total of approximately 82 dollars.
- D. 8 years after Immanuel purchased the rare coin, its predicted value is approximately 82 dollars.

A rectangle has a length that is 15 times its width. The function $y = 15ww$ represents this situation, where y is the area, in square feet, of the rectangle and $y > 0$. Which of the following is the best interpretation of $15w$ in this context?

- A. The length of the rectangle, in feet
- B. The area of the rectangle, in square feet
- C. The difference between the length and the width of the rectangle, in feet
- D. The width of the rectangle, in feet

A model predicts that the population of Bergen was 15,000 in 2005. The model also predicts that each year for the next 5 years, the population p increased by 4% of the previous year's population. Which equation best represents this model, where x is the number of years after 2005, for $x \leq 5$?

- A. $p = 0.9615,000^x$
- B. $p = 1.0415,000^x$
- C. $p = 15,0000.96^x$
- D. $p = 15,0001.04^x$

An egg is thrown from a rooftop. The equation $h = -4.9t^2 + 9t + 18$ represents this situation, where h is the height of the egg above the ground, in meters, t seconds after it is thrown. According to the equation, what is the height, in meters, from which the egg was thrown?

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.